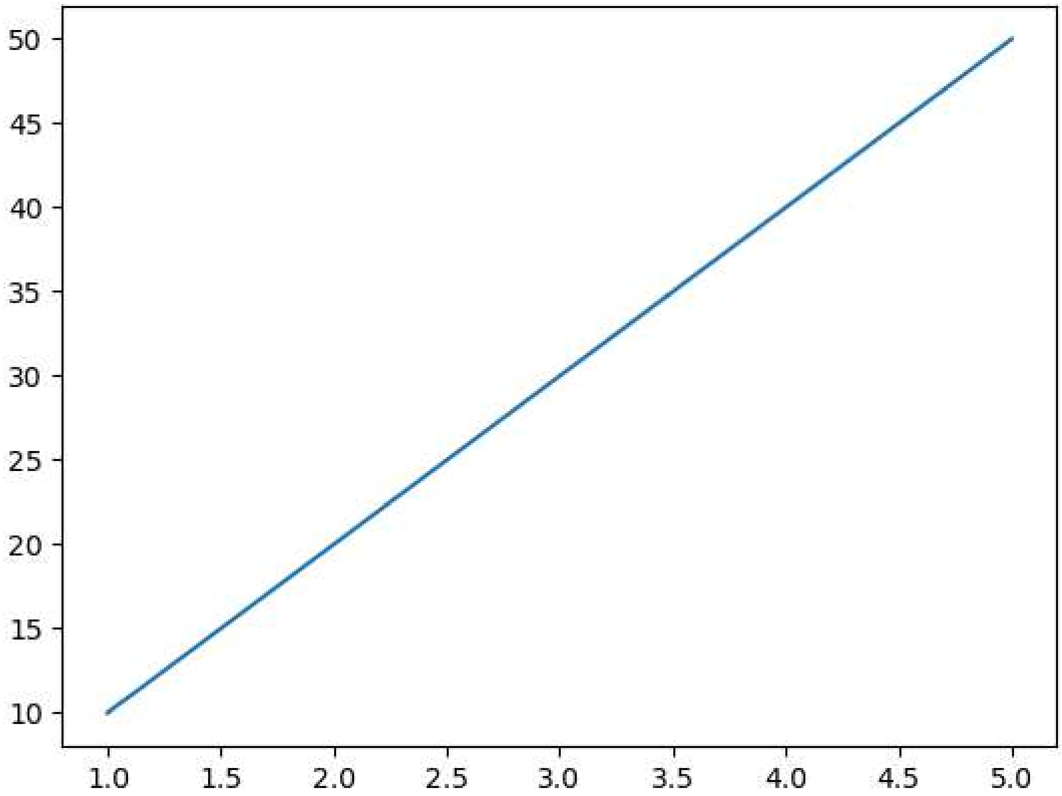
DATA VISUALIZATION

In [2]:
```
!pip install matplotlib
```

```
Requirement already satisfied: matplotlib in c:\users\admin\anaconda3\lib\site-packages (3.10.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (1.3.
3)
Requirement already satisfied: cycler>=0.10 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (4.6
0.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (1.
4.9)
Requirement already satisfied: numpy>=1.23 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (2.3.5)
Requirement already satisfied: packaging>=20.0 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib) (3.2.
5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib)
(2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\admin\anaconda3\lib\site-packages (from python-dateutil>=2.7->mat
plotlib) (1.17.0)
```

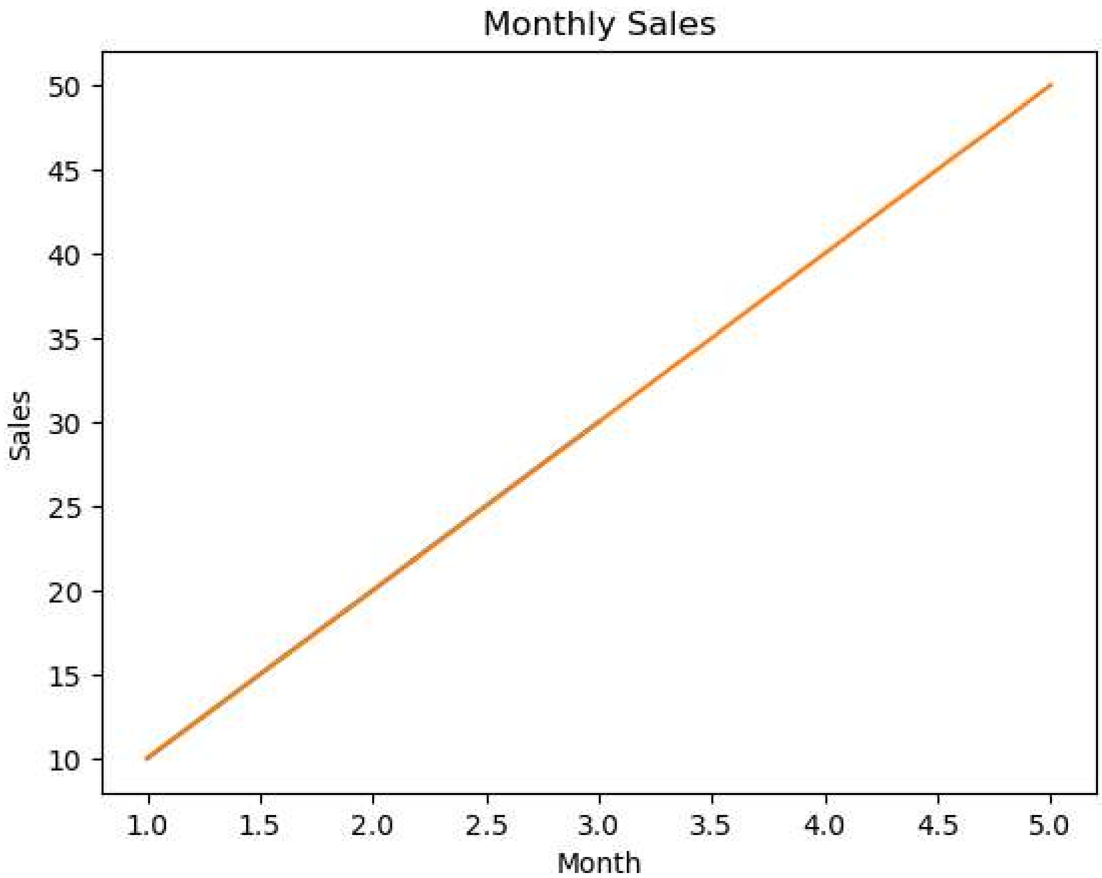
In [4]:
```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,30,40,50]
plt.plot(x,y)
plt.show()
```



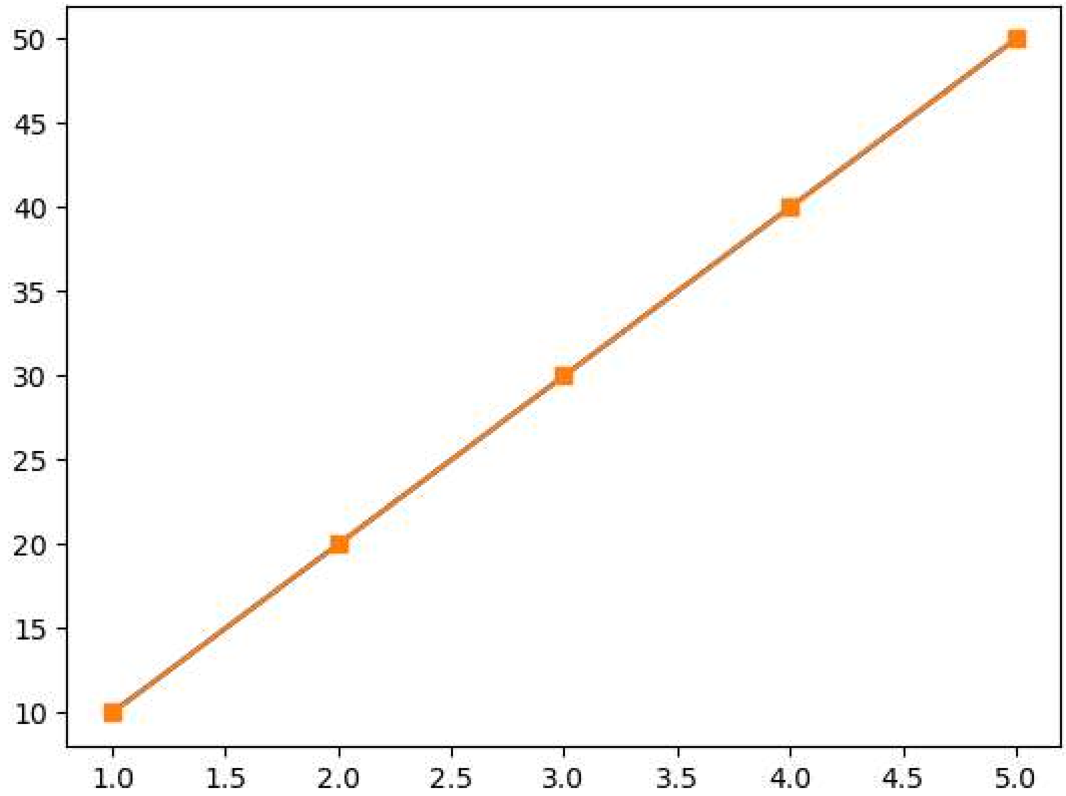
In [26]:

```
#figure and Axes
#There are must important concepts.
#Figure - The Entire Chart window.
#Axes - The Area where the group is drawn

import matplotlib.pyplot as plt
fig,ax = plt.subplots()
ax.plot([1,2,3],[10,20,30])

#Adding a Title
plt.title('Monthly Sales')

#Adding X and Y lables
plt.xlabel("Month")
plt.ylabel("Sales")
plt.plot(x,y)
```

```python
plt.show()

#Changing Line Color
plt.plot(x,y,color='blue')

#Changing line Style
plt.plot(x,y,linestyle = '-')

#Adding Markers
plt.plot(x,y,marker='s')

#o -circle
#s -Square
#^ -triangle
#* - Star
```



Out[26]: [<matplotlib.lines.Line2D at 0x1fd74402490>]



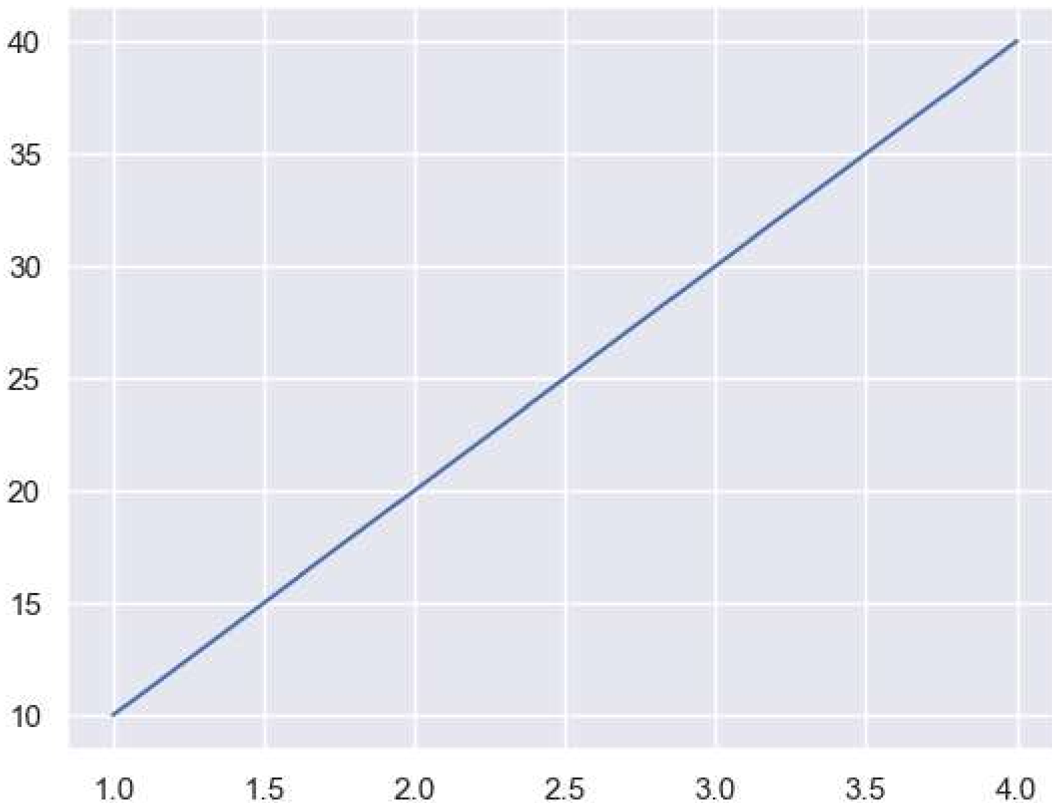
In [21]:
```
!pip install seaborn
```

```
Requirement already satisfied: seaborn in c:\users\admin\anaconda3\lib\site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\users\admin\anaconda3\lib\site-packages (from seaborn) (2.
3.5)
Requirement already satisfied: pandas>=1.2 in c:\users\admin\anaconda3\lib\site-packages (from seaborn) (2.3.3)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\users\admin\anaconda3\lib\site-packages (from seaborn)
(3.10.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=3.6.
1,>=3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=
3.4->seaborn) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=3.6.
1,>=3.4->seaborn) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=3.6.
1,>=3.4->seaborn) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=3.6.1,
>=3.4->seaborn) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4-
>seaborn) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=3.6.
1,>=3.4->seaborn) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\admin\anaconda3\lib\site-packages (from matplotlib!=
3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\admin\anaconda3\lib\site-packages (from pandas>=1.2->seaborn)
(2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\admin\anaconda3\lib\site-packages (from pandas>=1.2->seabor
n) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\admin\anaconda3\lib\site-packages (from python-dateutil>=2.7->mat
plotlib!=3.6.1,>=3.4->seaborn) (1.17.0)
```

In [33]:

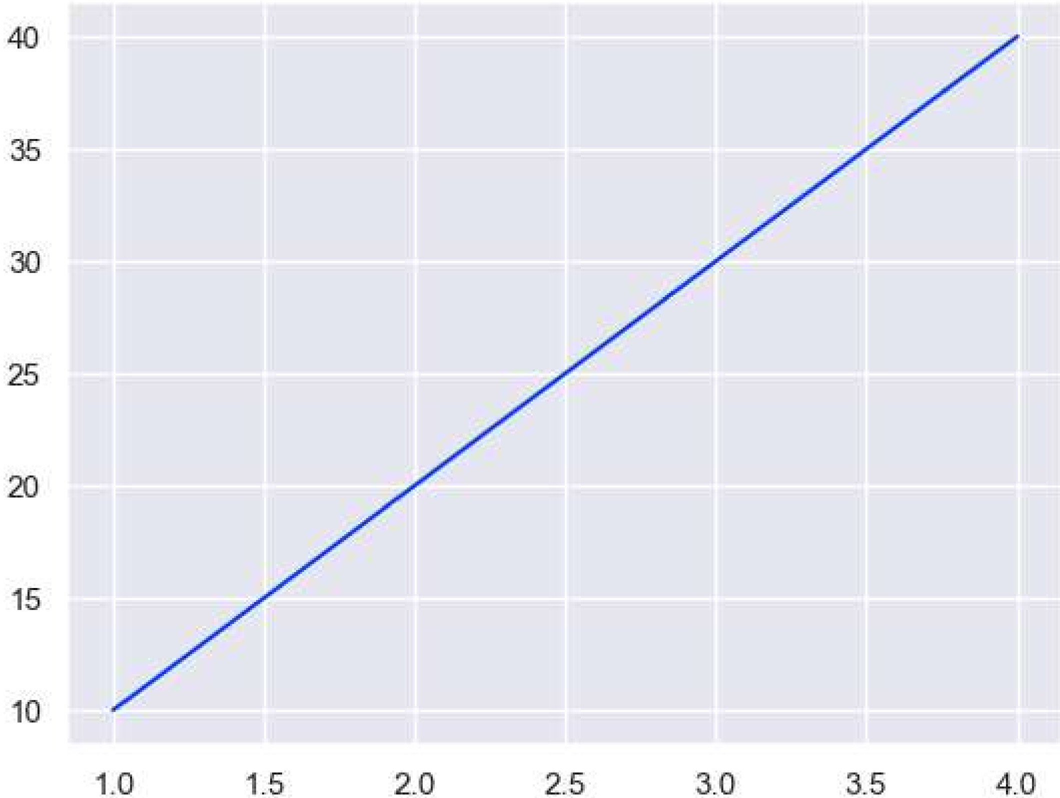
```python
import seaborn as sns
import matplotlib.pyplot as plt
x = [1,2,3,4]
y = [10,20,30,40]

#Themes in Seaborn
#Themes controd the appearence of charts
#Default Themes
sns.set_theme()

#Dark Grid Theme
sns.set_style('darkgrid')
```

```
#Available Styles
#DarkGrid
#White grid
#Dark
#white
sns.lineplot(x=x,y=y)
plt.show()

#A palattes is a collection of colors
#View Color Palatte
sns.set_palatte
```



In [47]:
```
#A palettes is a collection of colors
#View Color Palatte

#Available Colors
#deep
```

```
#Dark
#Bright
#Muted
#palette
#colorblind

sns.set_palette("bright")
sns.lineplot(
    x=[1,2,3,4],
    y=[10,20,30,40]
)
plt.show()
```



In [48]:

```
#Build in dataset support
#One of Seaborn's biggest advantages is build in datasets.
#view Available Dataset
```

```python
import seaborn as sns
print(sns.get_dataset_names())
```

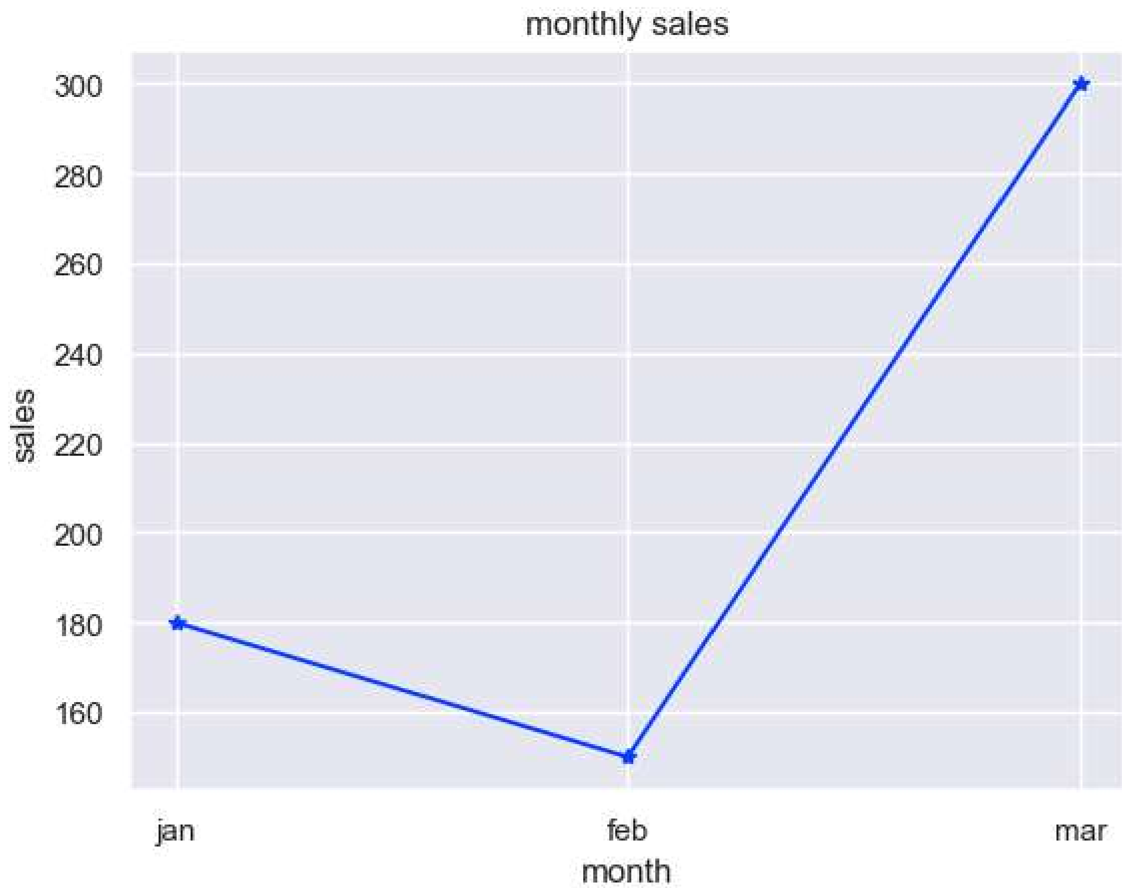
```
['anagrams', 'anscombe', 'attention', 'brain_networks', 'car_crashes', 'diamonds', 'dots', 'dowjones', 'exercise', 'f
lights', 'fmri', 'geyser', 'glue', 'healthexp', 'iris', 'mpg', 'penguins', 'planets', 'seaice', 'taxis', 'tips', 'tit
anic']
```

In [50]:
```python
#load titanic Dataset
import seaborn as sns
df = sns.load_dataset('titanic')
print(df.head())
```

```
   survived  pclass     sex   age  sibsp  parch     fare embarked  class  \
0         0       3    male  22.0      1      0   7.2500        S  Third
1         1       1  female  38.0      1      0  71.2833        C  First
2         1       3  female  26.0      0      0   7.9250        S  Third
3         1       1  female  35.0      1      0  53.1000        S  First
4         0       3    male  35.0      0      0   8.0500        S  Third

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True
```

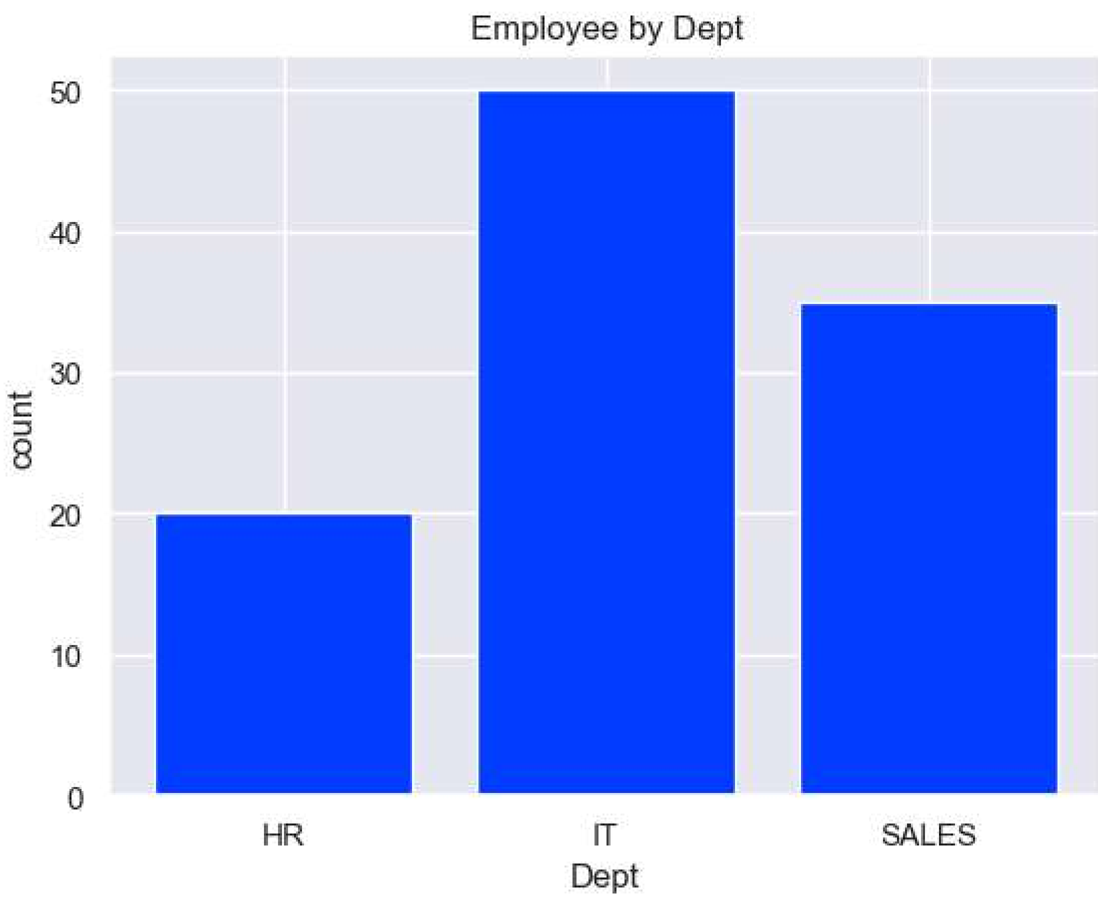
LINE CHART

In [58]:
```python
#line charts
#A line charts connect data points with lines
import matplotlib.pyplot as plt
month = ['jan','feb','mar']
sales=[180,150,300]

plt.plot(month,sales,marker='*')
plt.title("monthly sales")
plt.xlabel("month")
plt.ylabel("sales")
plt.show()
```



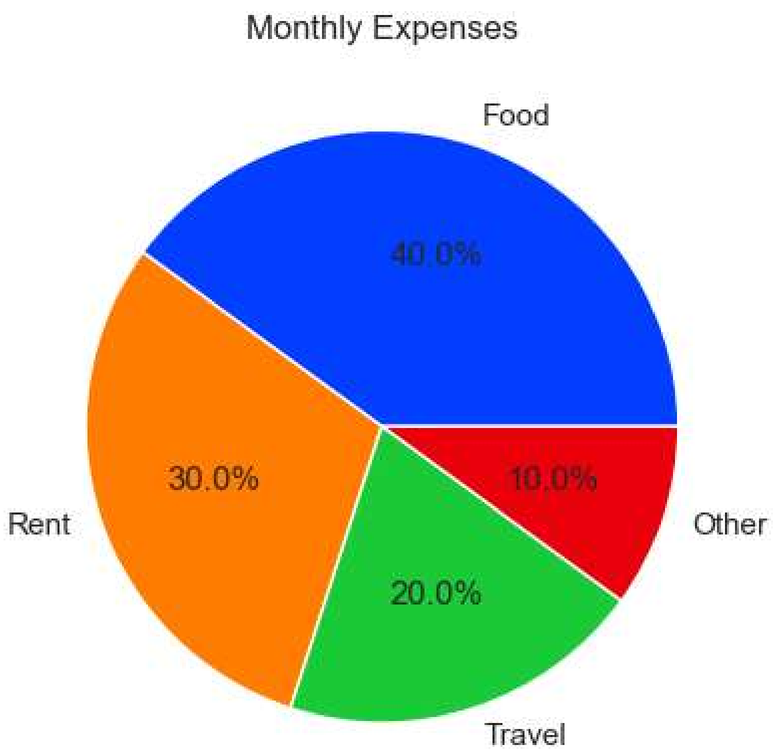
BAR CHART

In [57]:
```
#Compare Values
import matplotlib.pyplot as plt
dept=['HR','IT','SALES']
employees = [20,50,35]

plt.bar(dept,employees)
plt.title("Employee by Dept")
plt.xlabel("Dept")
plt.ylabel("count")
plt.show()
```



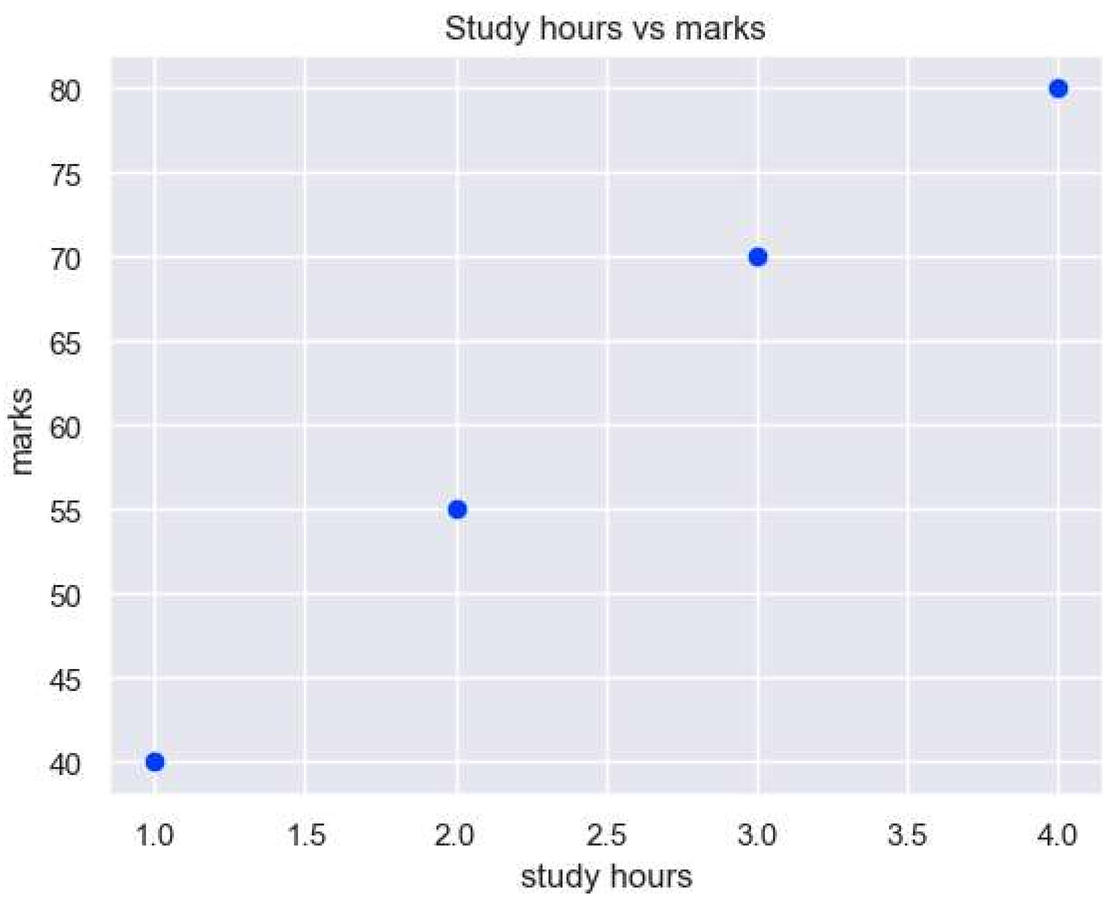
PIE CHART

In [64]:
```
# it shows part of a whole.
import matplotlib.pyplot as plt
labels = ['Food','Rent','Travel','Other']
values = [40,30,20,10]
plt.pie(values,labels = labels, autopct = "%1.1f%%")
plt.title("Monthly Expenses")
plt.show()
```



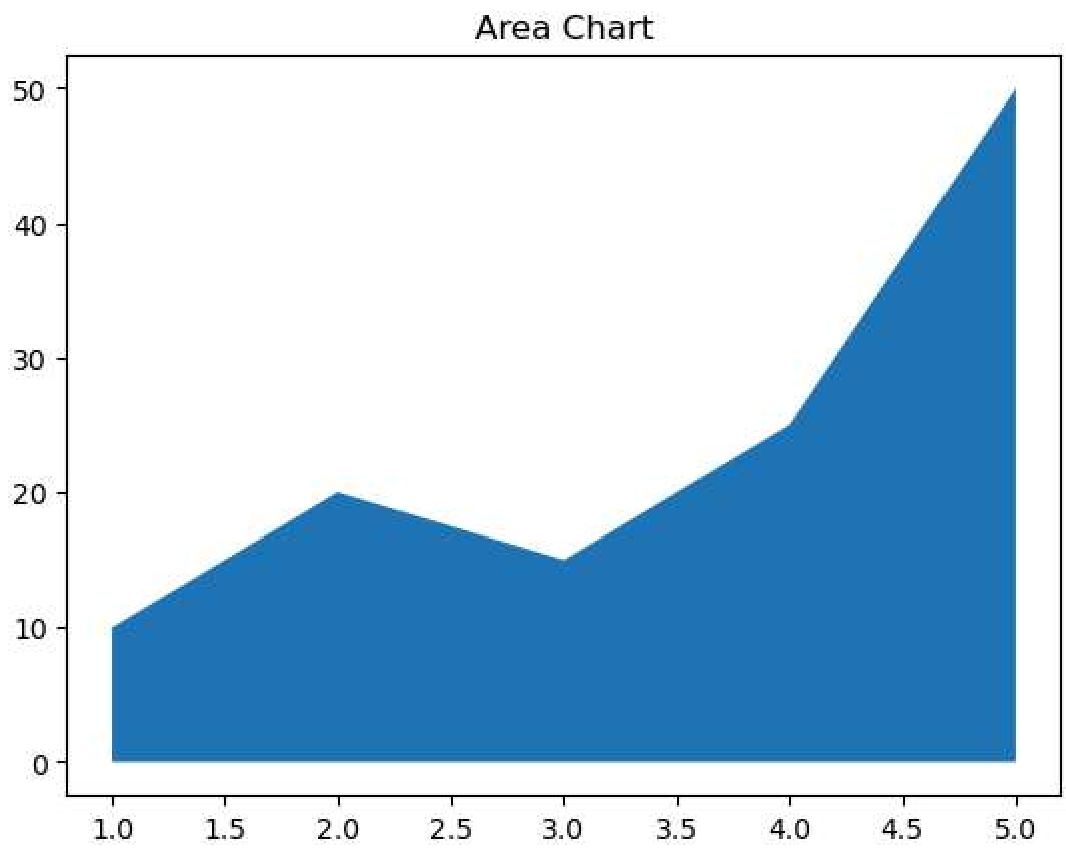
SCATTER PLOT

In [66]:
```
#Scatter plot
import matplotlib.pyplot as plt
hours=[1,2,3,4]
marks=[40,55,70,80]
plt.scatter(hours,marks)
plt.title("Study hours vs marks")
plt.xlabel("study hours")
plt.ylabel("marks")
plt.show()
```



AREA CHART

In [2]:
```
#Area chart
import matplotlib.pyplot as plt
x = [1,2,3,4,5]
y=[10,20,15,25,50]
plt.fill_between(x,y)
plt.title("Area Chart")
plt.show()
```



BOX PLOT

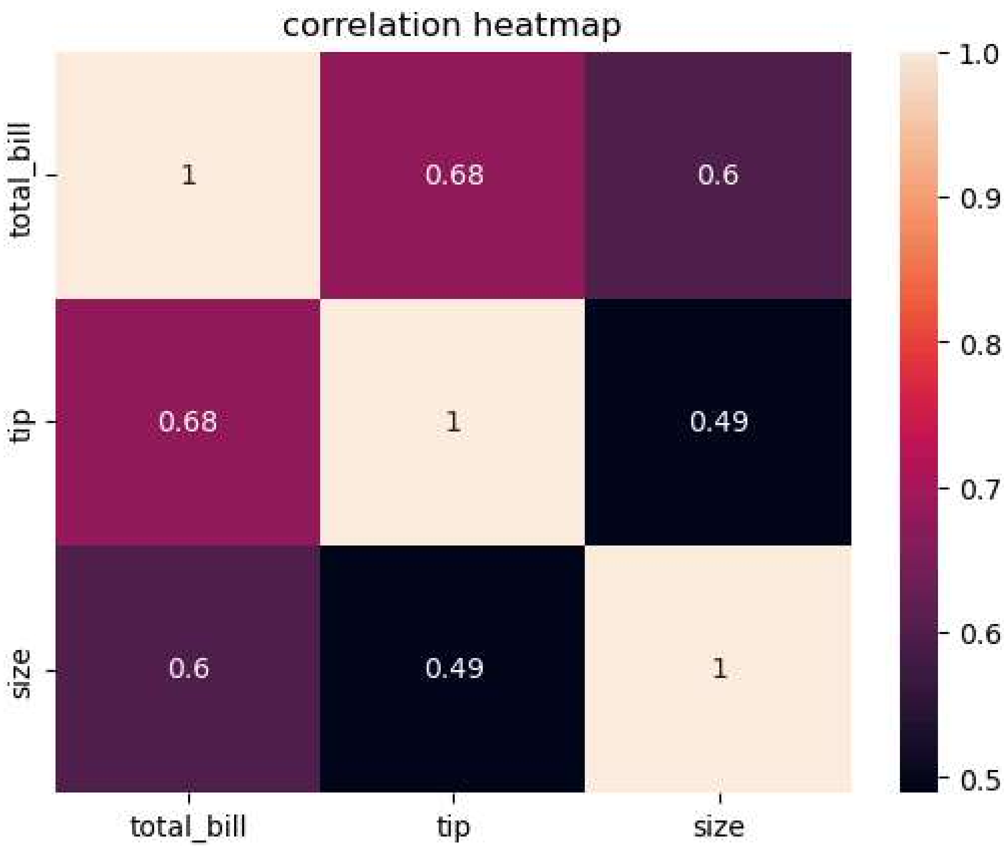
In [5]:

```
#Box plot is used to visualize the distribution of numrical data and identify outliers.
import matplotlib.pyplot as plt
salary =[20000,25000,30000,43000,56000,
         37000,26000,45000,26000,100000]
plt.boxplot(salary)
plt.title("Salary Distribution")
plt.xlabel("Salary")
plt.show()
```

HEAT MAP

In [8]:
```
#Heat map uses colors to represent values in a table o matrix.
#In data analysis heatmap are commonly used to visualize correlation between numerical column.

import matplotlib.pyplot as plt
import seaborn as sns

#Load Dataset
tips = sns.load_dataset("tips")

#Correlation matrix
corr = tips.corr(numeric_only=True)
```

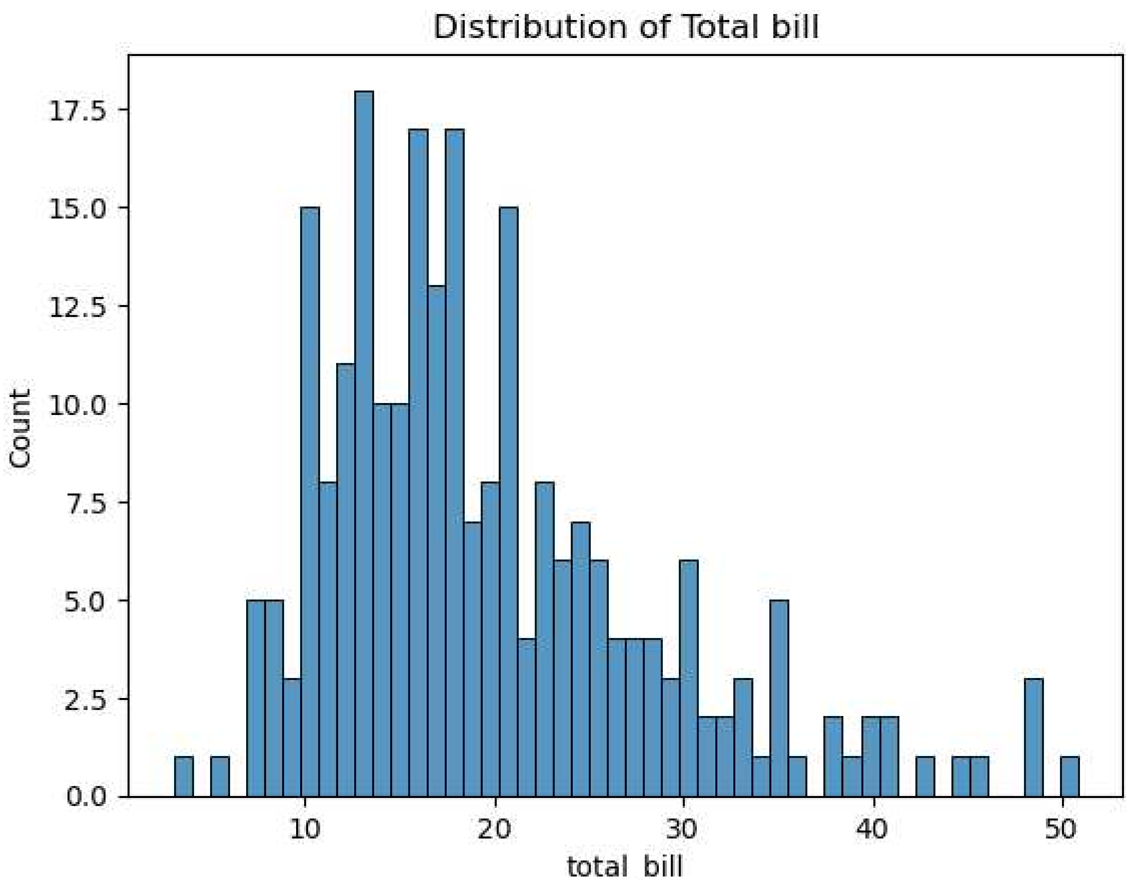
```python
#Heatmap
sns.heatmap(corr,annot=True)


plt.title("correlation heatmap")
plt.show()
```



Histogram

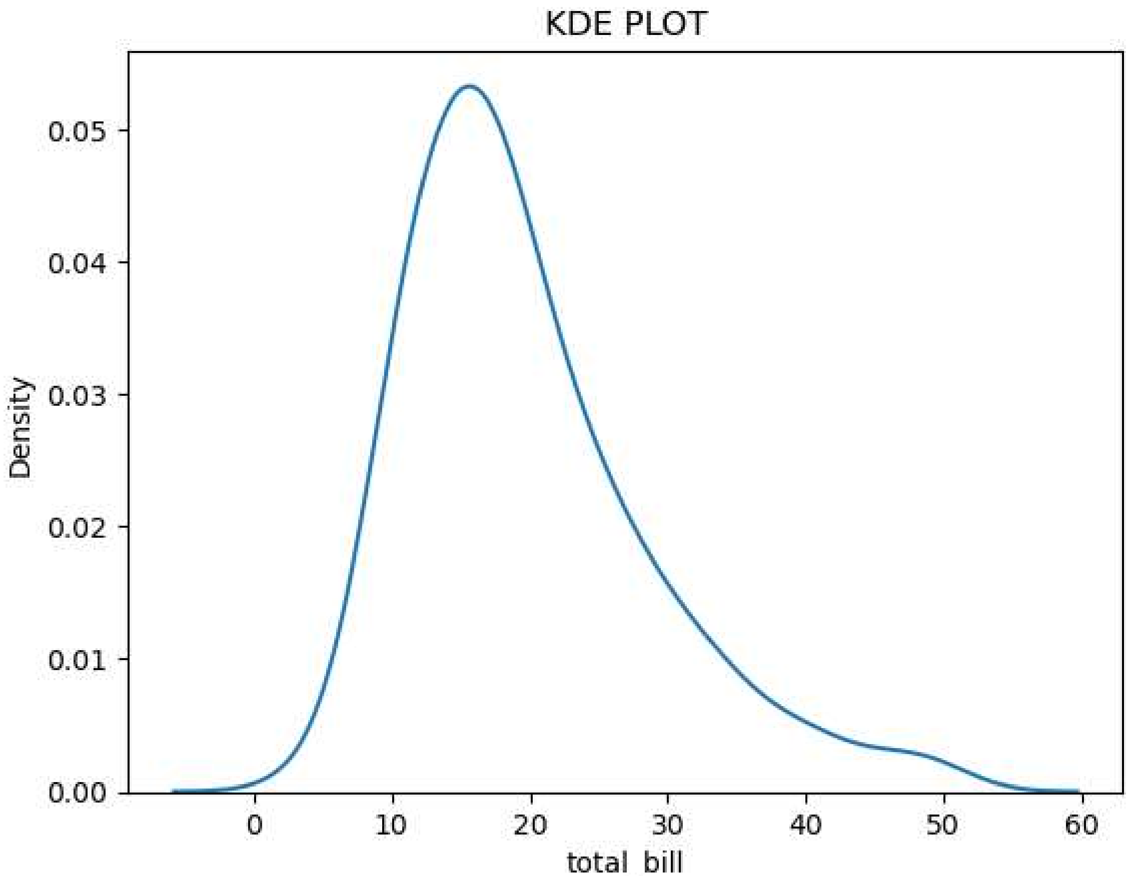
In [19]:
```python
#Histogram using seaborn
sns.histplot(data=tips,x="total_bill",bins=50)
plt.title("Distribution of Total bill")
plt.show()
```

```
#sns.histogram()
#Create Histogram.
```



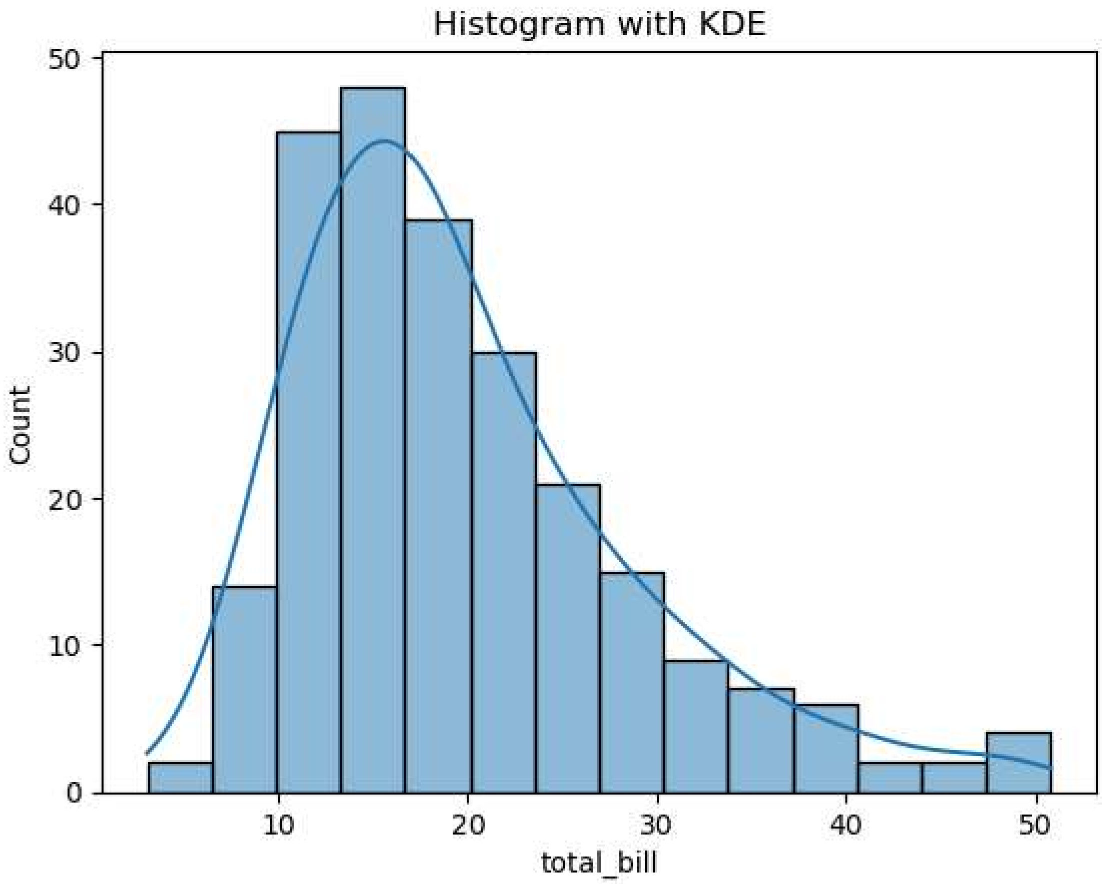
KDE PLOT

In [26]:
```
#Kernel Density Estimation
#smooth version of a histogram
#instead of bars,it shows a smooth curve representing data distribution.
sns.kdeplot(data=tips,x="total_bill")
plt.title("KDE PLOT")
plt.show()
```



HISTOGRAM WITH KDE

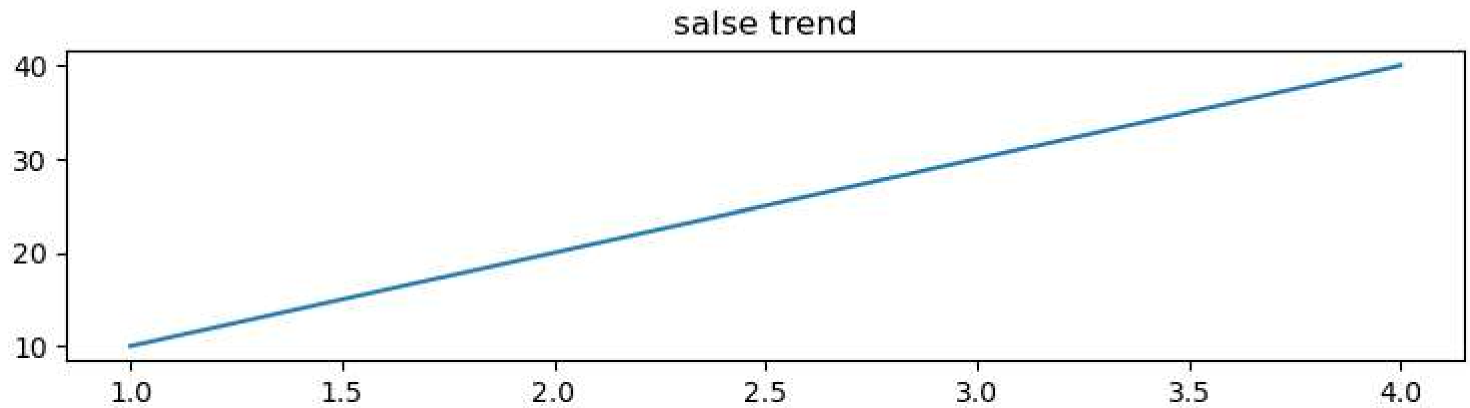
In [27]:
```
#Histogram with KDE
sns.histplot(data=tips,x="total_bill",kde=True)
plt.title("Histogram with KDE")
plt.figure(figsize=(9,2))
plt.show()
```




```
<Figure size 900x200 with 0 Axes>
```

In [32]:

```
#Figure size controls the width and height of a chart
import matplotlib.pyplot as plt
plt.figure(figsize=(9,2))
plt.plot([1,2,3,4],[10,20,30,40])
plt.title("salse trend")
plt.savefig("Sales_chart.png",dpi=100)
plt.show()
```



SUBPLOTS

In [ ]:
```
#Subplots allow multiple charts in a single figure.
```